

Node features (d = 10)

Symbol	Unit	Description	Source
x	m	Joint x-coordinate	geom
y	m	Joint y-coordinate	geom
z	m	Joint z-coordinate	geom
T_x	—	Fixed-DOF flag, translate x	BC
T_y	—	Fixed-DOF flag, translate y	BC
T_z	—	Fixed-DOF flag, translate z	BC
R_x	—	Fixed-DOF flag, rotate x	BC
R_y	—	Fixed-DOF flag, rotate y	BC
R_z	—	Fixed-DOF flag, rotate z	BC
F_z	N	Nodal load (tributary area)	load

Edge features (d = 9)

Symbol	Unit	Description	Source
b	m	Section width	stock
h	m	Section depth	stock
L	m	Installed member length	geom
E	N m^{-2}	Elastic modulus	stock
I_y	m^4	2nd moment of area (y -axis)	stock
I_z	m^4	2nd moment of area (z -axis)	stock
J	m^4	Torsional constant (St. Venant)	stock
EA/L	N m^{-1}	Axial stiffness	geom
N_{EA}	N	Mean-EA axial force estimate	geom

geometry-derived

boundary condition

applied load

stock property